

Riya Chamala

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EDUCATION

Purdue University

B.S.E in Computer Engineering, Minor in Cybersecurity

West Lafayette, IN

Aug. 2023 – May 2026

- Software and Tech Consultant, Project Manager @ Purdue Solutions Consulting
- Technological Investment Analyst @ Purdue Investment and Trading (ITP)
- Computer Engineering and Program Mentor @ Women in Engineering at Purdue

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS, MATLAB, R

Frameworks and Libraries: React.js, PyTest, Pandas, TensorFlow, PyTorch, Scikit-learn, NumPy, Flask, Matplotlib

Developer Tools: GitHub, PowerBI, Docker, RESTful API, AWS, Azure, VS Code, PyCharm, IntelliJ, Vocareum

Certifications: AWS Cloud Practitioner (WIP), Azure Developer Associate (WIP)

PROFESSIONAL EXPERIENCE

Engineering Intern & Lab Technician

May 2024 – Aug. 2024

Samtec Inc.

New Albany, IN

- Engineered robust software modules and APIs in Python and MATLAB to modernize legacy wiring analysis and structural integrity testing, incorporating optimization algorithms and scalable architectures to enhance system performance.
- Developed advanced automation frameworks that orchestrated efficient API calls across diverse technology stacks, enabling real-time data acquisition from measurement devices and optimizing analytics processes.
- Spearheaded comprehensive process documentation and cross-functional collaboration to institutionalize best practices, fostering a culture of continuous improvement and technical excellence.

Software Engineering Consultant - Internal Operations Restructuring

Jan. 2024 – May 2024

Purdue Solutions Consulting

Chicago, IL

- Designed a revenue-optimization algorithm using Python and Pandas to restructure internal payment systems, improving operational efficiency while integrating robust encryption protocols and secure SQL schema designs to protect financial data.
- Developed and integrated performance tracking and visualization components with Matplotlib and Power BI, aligning enhancements with project goals and conducting regular vulnerability assessments to ensure system integrity.
- Collaborated with stakeholders to refine agile processes by integrating backlog grooming and story sizing into the development lifecycle, ensuring scalable technical improvements and transparent project execution.

Software Engineer - Real Estate Data Management System

Aug. 2023 – Dec. 2023

Purdue University

West Lafayette, IN

- Developed and deployed a robust Zillow API integration that aggregated data from over 5,000 property listings, streamlining data ingestion for dynamic applications and implementing secure API communication protocols to safeguard data integrity.
- Designed and implemented a responsive, interactive UI using React.js that supported scalable growth from 500 to 1,000 monthly active users through efficient backend API integration and secure data exchange practices.
- Engineered advanced data filtering and sorting mechanisms with Pandas and NumPy, optimizing query execution across 2,000+ listings while integrating encryption and periodic vulnerability assessments to protect sensitive market data.

Primary Researcher - Bioinformatics Machine Learning Project

Sept. 2022 – Feb 2023

Academies of Loudoun & Griffith University

Leesburg, VA

- Architected and developed a scalable Python-based machine learning pipeline for sensitive bioinformatics data, integrating encryption protocols and designing secure SQL schemas for optimized data storage.
- Engineered a TensorFlow CNN with secure coding practices, logging, and anomaly detection, incorporating parameterized SQL queries to mitigate injection risks and achieve an AUC-ROC of 0.92.
- Collaborated with cross-functional teams to integrate security and quality testing into the lifecycle through code reviews, security workshops, and periodic SQL vulnerability assessments.